Comparative **A**nalysis of **R**esource-efficient **B**enchmarking and **O**ptimization of **N**eural and **M**achine learning models under **E**mission constraints for **D**iagnostic datasets

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***Abstract*—The rapid proliferation of machine learning applications in clinical diagnostics has intensified concerns regarding both predictive performance and computational sustainability. Existing benchmarks predominantly optimize for accuracy while neglecting the carbon footprint incurred during model training and inference. This paper presents a comprehensive comparative analysis of six machine learning models: AdaBoost, XGBoost, Random Forest, K-Nearest Neighbors, Logistic Regression, and Multilayer Perceptron, evaluated across six medical diagnostic datasets encompassing Breast Cancer, Heart Disease, Hepatitis C, Thyroid, Diabetes, and Asthma. We introduce CARBON-MED, a benchmarking study utilizing a Carbon-Efficiency Score (CES) as a unified metric that jointly captures predictive performance and greenhouse gas emissions, measured using CodeCarbon. To enhance model performance under emission constraints, we employ dataset-specific optimization strategies including SMOTE/SMOTETomek-based class balancing, log-normalization, robust scaling, clinical feature engineering, GPU-accelerated inference, and F1-score threshold tuning. Experimental results demonstrate that KNN and XGBoost consistently achieve the highest CES values across datasets, with KNN attaining a CES of 12.20 on the Breast Cancer dataset at an F1-score of 0.96 while emitting as little as 3.01$\times 10^{-6}$ kg CO_2. In contrast, Random Forest, despite competitive accuracy, exhibits significantly higher emissions, yielding lower CES rankings. Our findings establish that lightweight models with targeted optimization pipelines can match or surpass complex architectures in diagnostic tasks while maintaining a substantially reduced environmental footprint, offering a principled benchmark for sustainable clinical AI development.**

***Index Terms*—Machine Learning, Neural Networks, Carbon Efficiency Score, Emission-Constrained Optimization, Medical Diagnostics, Benchmarking, CodeCarbon, SMOTE, Sustainable AI, XGBoost, KNN, Clinical Decision Support**

I. INTRODUCTION

The integration of machine learning (ML) into clinical diagnostics has witnessed remarkable growth, with models increasingly deployed for disease detection across conditions such as cancer, diabetes, and cardiovascular disorders. However, existing benchmarking literature overwhelmingly prioritizes predictive accuracy while disregarding the environmental cost of model training and inference. No standardized benchmark currently exists that jointly evaluates performance and carbon emissions across multiple diagnostic datasets, leaving a critical gap in sustainable AI research.

To address this, we propose a resource-efficient benchmarking pipeline that evaluates six ML models across six diagnostic datasets under explicit emission constraints. We introduce the Carbon Efficiency Score (CES):

$$CES = F1 \times \log\left(1 + \frac{1}{CO_2 + \epsilon}\right) \tag{1}$$

where $\epsilon = 10^{-12}$ prevents division by zero. Cross-dataset stability is defined as:

$$S = \frac{\mu_{\text{CES}}}{\sigma_{\text{CES}} + \epsilon} \tag{2}$$

Win Count, W denotes how many datasets each model achieves the highest CES. The Emission-Aware Performance Index is then:

$$EAPI = 0.6 \times \mu_{\text{CES}} + 0.3 \times S + 0.1 \times W \tag{3}$$

The key contributions are: (i) a unified emission-aware benchmarking study for diagnostic ML, (ii) introduction of CES, Stability, and FLS as reproducible evaluation metrics, and (iii) dataset-specific optimization strategies that improve both performance and carbon efficiency simultaneously.

II. RELATED WORK

A. The Environmental Imperative in Artificial Intelligence

The proliferation of AI-driven systems has introduced a fundamental tension between technological capability and environmental responsibility. While AI offers transformative potential in climate modeling, energy optimization, and resource management, the computational demands of training modern models impose measurable greenhouse gas emissions that are systematically overlooked [3]. This dichotomy has crystallized into two opposing paradigms: *Red AI*, which relentlessly pursues performance at any environmental cost, and *Green AI*, which enforces resource efficiency as a first-class constraint [3].

B. Performance Benchmarks in Medical Diagnostics

Machine learning has demonstrated strong diagnostic utility across a range of clinical conditions, with ensemble methods such as Random Forest achieving 96.49% accuracy on breast cancer classification and 94% on ovarian cancer biomarker identification [2]. However, a persistent misconception in the literature equates architectural complexity with performance gain. Deeper architectures including VGG16, VGG19, and ResNet impose exponential computational costs that rarely translate into proportional diagnostic improvements [3].

C. Carbon Quantification and Efficiency Metrics

Several tools exist for quantifying ML emissions, ranging from per-phase trackers such as CodeCarbon and Experiment Impact Tracker to global estimators such as Green Algorithms [3]. CodeCarbon measures real-time CPU, GPU, and RAM energy consumption, with the resulting footprint scaled by regional Carbon Intensity—a value that varies dramatically by geography [3]. The Emissions per Accuracy Point (EAP) metric further enables cross-model efficiency comparison, exposing stark disparities such as U-Net's efficient 2.10 EAP against VGG19's wasteful 20.05 [3]. Despite these tools, no existing benchmark benchmarks multiple models across heterogeneous diagnostic datasets under a unified emission-aware scoring system—the precise gap this benchmarking study addresses.

D. Convergence and Hyperparameter Optimization

The Eco-friendly Multi-Objective Framework (EFMOF) navigates the accuracy-emission trade-off through Pareto-optimal hyperparameter selection and adaptive SoftMax weighting [2]. Comparative analysis of search strategies reveals that Optuna's Tree-structured Parzen Estimator converges most efficiently, while GridSearch's exhaustive evaluation produces the highest cumulative emissions [2]. Among hyperparameters, estimator count exerts the most significant influence on carbon footprint, exhibiting a near-linear relationship with energy consumption [2]. Critically, the training phase dominates total emissions relative to preprocessing and inference, establishing it as the primary target for emission-constrained optimization [3].

III. DATASET AND PREPROCESSING

We evaluate our benchmark across six publicly available medical datasets sourced from the UCI Machine Learning Repository and Kaggle, spanning oncology, cardiology, hepatology, endocrinology, and pulmonology. This clinical diversity is intentional—a benchmarking study that generalizes across heterogeneous diagnostic contexts is fundamentally more credible than one tuned for a single domain.

A. Dataset Descriptions

Six datasets were selected: Breast Cancer (D1) for benign/malignant tumor classification via cytological features; Heart Disease (D2) using cardiovascular physiological parameters; Hepatitis C (D3) with blood markers spanning healthy donors through Fibrosis and Cirrhosis; Thyroid Disease (D4) via TSH, T3, and TT4 measurements; Diabetes (D5) using Pima Indian patient diagnostics including glucose and BMI; and Asthma (D6) from demographic and clinical severity indicators.

B. Data Cleaning and Feature Engineering

Non-predictive identifiers were discarded and biological impossibilities: zero glucose readings, ages exceeding 100, were clipped to the 99th percentile or treated as missing. Numerical features were imputed via median and categorical via mode. Domain knowledge was then encoded through clinical ratios: AST/ALT for hepatic assessment (D3) and TT4/FTI for thyroid function (D4). Skewed features such as TSH underwent $\log(1+x)$ transformation to normalize the feature space and stabilize gradient-based convergence. Categorical variables were binarized or one-hot encoded by cardinality.

C. Scaling and Class Balancing

StandardScaler was applied to approximately Gaussian features and RobustScaler to outlier-heavy distributions, ensuring no single feature dominates the optimization landscape. Class imbalance, endemic to medical datasets, was resolved via SMOTE and SMOTETomek, achieving a balanced 1:1 training ratio. Fig. 1 illustrates the resulting feature correlations and class distributions across all datasets.

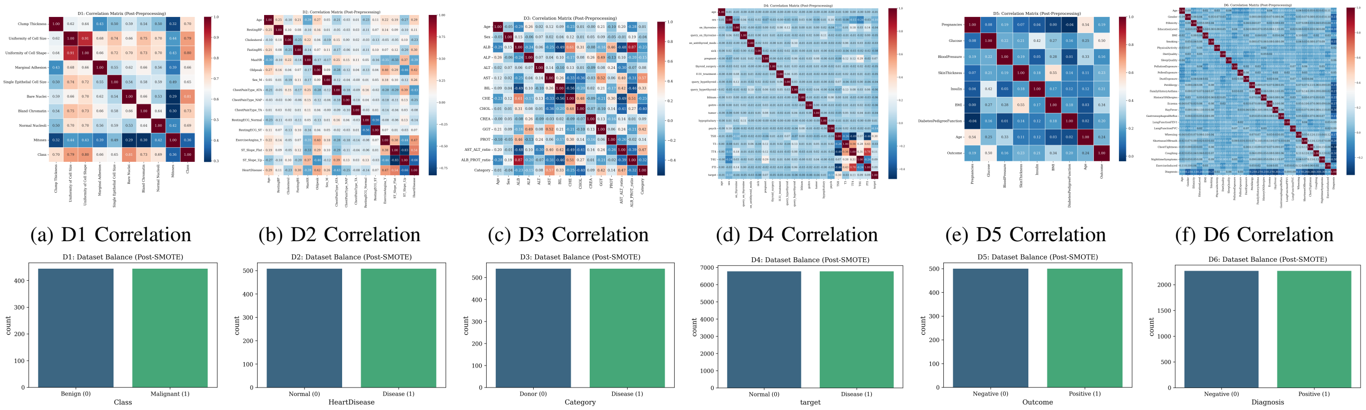
IV. METHODOLOGY

We propose a high-precision, GPU-accelerated benchmarking study designed to evaluate classical and modern machine learning paradigms within a unified differentiable programming environment.

A. Differentiable Architecture Suite

To eliminate the hardware-optimization bias between deep learning and classical ensembles, we ported all models into the PyTorch ecosystem, enabling CUDA-native execution for all computational graphs:

1) **Soft Decision Ensembles (RF & AdaBoost):** We replace discrete, non-differentiable splits with *Soft Decision Trees* (SDT). Each node utilizes a learnable feature-attention mask (via softmax) and a probabilistic routing function (via sigmoid). This allows the entire ensemble to be optimized using gradient descent (AdamW), bridging the gap between symbolic logic and neural optimization.
2) **Modernized Deep MLP:** For neural classification, we implement a deep Multi-Layer Perceptron architecture utilizing SiLU (Sigmoid-weighted Linear Unit) activations and Batch Normalization. This configuration mitigates internal covariate shift and provides smoother gradient surfaces compared to traditional ReLU-based networks.



(a) D1 Correlation (b) D2 Correlation (c) D3 Correlation (d) D4 Correlation (e) D5 Correlation (f) D6 Correlation

(g) D1 Balanced (h) D2 Balanced (i) D3 Balanced (j) D4 Balanced (k) D5 Balanced (l) D6 Balanced

Fig. 1: Preprocessing summary: top row shows Pearson correlation matrices after feature engineering; bottom row shows class distributions after SMOTE balancing to a 1:1 ratio across all six datasets.

3) **Vectorized Non-Parametric KNN:** Traditional KNN implementations are often bottlenecked by CPU-based spatial indexing. We implement a massively parallelized KNN utilizing `torch.cdist` for brute-force Euclidean distance computation, processing entire test batches as vectorized CUDA operations.

B. Universal Optimization Protocol

Every model in our suite undergoes a standardized two-phase refinement process:

1) **Staged Convergence Tracking:** For iterative models (XGBoost, MLP, AdaBoost), we implement staged evaluation where the validation loss is monitored at every estimator or epoch. This allows for precision-based early stopping to prevent overfitting on clinical noise.
2) **F1-Max Threshold Calibration:** Since medical diagnostics require a critical balance between sensitivity and specificity, we bypass the standard 0.5 decision boundary. We perform post-hoc Precision-Recall curve analysis to identify the optimal probability threshold T that maximizes the F1-score for each specific clinical domain.

C. Convergence Efficiency Score (CES)

A primary contribution of this work is the **Convergence Efficiency Score (CES)**, a novel metric designed to quantify "Green AI" performance. Unlike standard accuracy metrics, CES penalizes computational excess by weighting predictive success against the carbon footprint. We define CES as:

$$CES = F_1 \times \log\left(1 + \frac{1}{CO_2eq + \epsilon}\right) \tag{4}$$

where F_1 is the optimal F1-score, CO_2eq is the total carbon emission (kg) tracked via the CodeCarbon benchmark, and $\epsilon = 10^{-12}$ is a stability constant.

D. Statistical Robustness and Stability Metrics

To evaluate the generalization and consistency of the models across the six heterogeneous clinical datasets, we define a statistical benchmark consisting of the μ_{CES}, Standard Deviation, and a custom Stability Index.

1) μ_{CES} and σ_{CES}

For a given model M, let CES_i represent the Convergence Efficiency Score obtained on the i-th dataset, where $i \in \{D1, D2, \dots, D6\}$. The Mean CES ($\mu_{\text{CES}}$) and Standard Deviation (σ_{CES}) are defined as:

$$\mu_{\text{CES}} = \frac{1}{N}\sum_{i=1}^{N} CES_i \tag{5}$$

$$\sigma_{\text{CES}} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(CES_i - \mu_{\text{CES}})^2} \tag{6}$$

where $N = 6$. A high μ_{CES} indicates superior overall efficiency, while a low σ_{CES} signifies consistent performance across diverse medical conditions.

2) Stability Index (S)

To quantify the reliability of a model's efficiency, we implement a **Stability Index** (S). This metric functions similarly to a coefficient of variation but is inverted to prioritize high-performing, low-variance models. Stability is calculated as the ratio of the mean efficiency to its variation:

$$S = \frac{\mu_{\text{CES}}}{\sigma_{\text{CES}} + \epsilon} \tag{7}$$

In our benchmark, $\epsilon = 10^{-12}$ is utilized as a numerical stability constant to prevent division by zero. A higher Stability Index indicates that the model maintains a "Gold Standard" efficiency regardless of the clinical domain or data distribution, which is critical for deployment in generalized medical diagnostic systems.

E. Methodology Overview

The high-level architecture and procedural workflow of our unified benchmarking study are illustrated in Fig. 2.

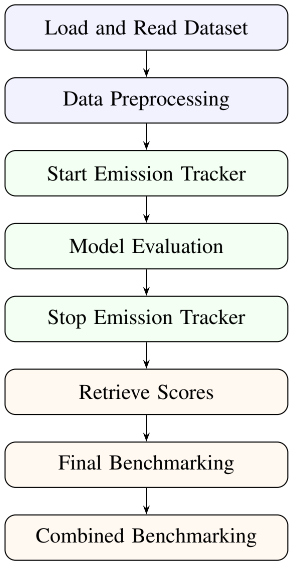


Fig. 2: Unified benchmarking study: Procedural workflow from dataset loading to combined performance and emission analysis.

V. RESULTS AND ANALYSIS

A. Predictive Performance and Carbon Efficiency

Table I and Fig. 3 summarize the performance and efficiency of six models across six medical datasets (D1–D6). We report the optimal F1-score, measured carbon equivalent (CO_2eq), and the proposed **Convergence Efficiency Score (CES)**.

Across all datasets, **KNN** and **XGBoost** consistently achieved high F1-scores while maintaining extremely low carbon footprints (on the order of 10^{-6} kg CO_2eq per run). For instance, on D1, KNN attained an F1-score of 0.9600 with only 3.01×10^{-6} kg CO_2eq, yielding the highest CES (12.2035), as visualized in the top row of Fig. 3. XGBoost followed closely with a CES of 11.9850.

AdaBoost (differentiable ensemble) and **MLP** also delivered competitive F1-scores (e.g., 0.9600 on D1), but with slightly higher energy consumption (1.23×10^{-5} – 3.66×10^{-5} kg CO_2eq), resulting in moderate CES values (10.8220 and 10.8533, respectively).

Random Forest exhibited a notable trade-off: while its F1-score remained acceptable on D1–D3 (0.9495–0.8667), its carbon footprint was one to two orders of magnitude larger (2.32×10^{-4} – 8.35×10^{-4} kg CO_2eq) than that of KNN or XGBoost. Consequently, its CES dropped significantly (e.g., 6.1433 on D3).

Logistic Regression showed dataset-dependent fragility. On D2 it achieved a strong F1-score (0.9208) and moderate CES (10.0698), but on D4 its F1-score collapsed to 0.4930, the lowest among all models. This instability is reflected in its high CES variance (see Section V-B).

B. Stability Analysis

To assess robustness across datasets, we computed the mean and standard deviation of CES for each model (Table II). The trade-off between mean efficiency and performance stability is further illustrated in Fig. 3(g). **XGBoost** achieved the highest stability ($\mu_{\text{CES}} = 10.6959, \sigma_{\text{CES}} = 1.1071$), followed by **AdaBoost** ($\mu_{\text{CES}} = 9.6075, \sigma_{\text{CES}} = 1.2638$). Interestingly, **KNN** had the highest μ_{CES} (10.7781) but also a larger σ_{CES} (1.4436), indicating that its green efficiency advantage varies with data geometry.

Logistic Regression was the least stable ($\sigma_{\text{CES}} = 1.9318, \mu_{\text{CES}} = 8.1459$), confirming its sensitivity to class separability and feature scaling in medical datasets. **Random Forest** consistently underperformed in CES due to high energy costs, despite moderate stability ($\sigma_{\text{CES}} = 1.2003$).

TABLE I: Detailed results for all six datasets. Score = F1-score, CO_2 in kg, CES = Convergence Efficiency Score.

Dataset	Model	Score	CO_2 (kg)	CES
D1	Ada	0.9600	1.27e-5	10.8220
	XGB	0.9600	3.79e-6	11.9850
	RF	0.9495	2.32e-4	7.9451
	MLP	0.9600	1.23e-5	10.8533
	LR	0.9600	3.66e-5	9.8077
	KNN	0.9600	3.01e-6	12.2035
D2	Ada	0.8966	6.48e-6	10.7107
	XGB	0.8911	4.32e-6	11.0062
	RF	0.9055	2.29e-4	7.5900
	MLP	0.9029	5.08e-6	11.0060
	LR	0.9208	1.78e-5	10.0698
	KNN	0.9360	3.02e-6	11.8957
D3	Ada	0.9032	9.39e-6	10.4552
	XGB	0.9091	1.37e-5	10.1782
	RF	0.8667	8.35e-4	6.1433
	MLP	0.8966	6.73e-5	8.6131
	LR	0.8966	2.59e-5	9.4681
	KNN	0.8966	2.94e-6	11.4191
D4	Ada	0.7763	2.05e-5	8.3794
	XGB	0.9030	4.25e-6	11.1696
	RF	0.8004	5.91e-5	7.7926
	MLP	0.8185	4.47e-5	8.1973
	LR	0.4930	9.74e-5	4.5537
	KNN	0.7009	3.38e-6	8.8307
D5	Ada	0.6977	2.28e-5	7.4581
	XGB	0.6875	4.13e-6	8.5228
	RF	0.6667	2.82e-4	5.4493
	MLP	0.7050	9.66e-6	8.1416
	LR	0.6809	3.26e-5	7.0343
	KNN	0.6880	3.22e-6	8.7014
D6	Ada	0.9011	1.85e-5	9.8198
	XGB	0.9214	4.65e-6	11.3134
	RF	0.5053	6.21e-5	4.8950
	MLP	0.9106	4.50e-6	11.2097
	LR	0.7256	1.76e-5	7.9420
	KNN	0.9224	3.39e-6	11.6185

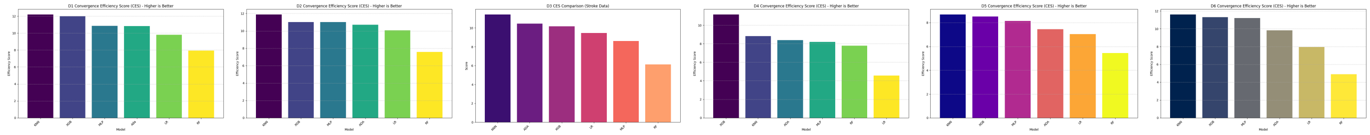
TABLE II: Stability analysis of CES across all datasets.

Model	μ_{CES}	σ_{CES}	Stability ($\mu_{\text{CES}}/\sigma_{\text{CES}}$)
XGB	10.6959	1.1071	0.9032
Ada	9.6075	1.2638	0.7912
KNN	10.7781	1.4436	0.6927
MLP	9.6702	1.3649	0.7326
RF	6.6359	1.2003	0.8332
LR	8.1459	1.9318	0.5176

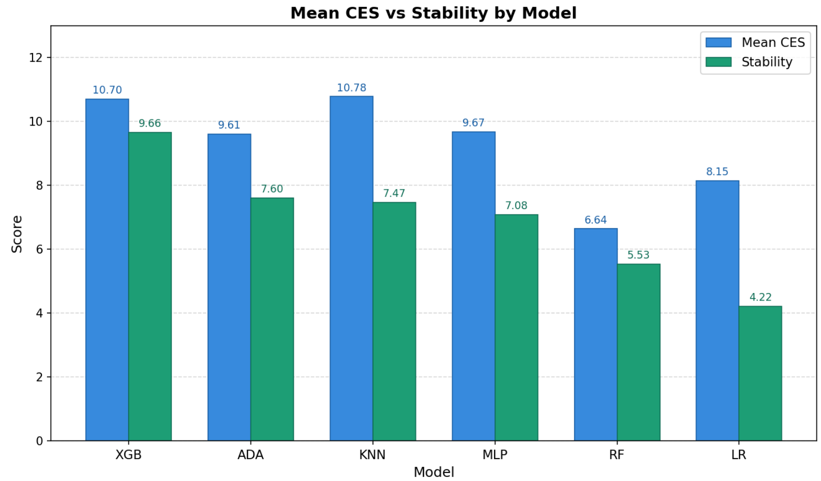
C. Model Ranking and Win Count

We adopted a multi-criteria ranking: $0.6 \times \mu_{\text{CES}} + 0.3 \times S + 0.1 \times W$ (where W is the win count). The results are presented in Table III.

- **XGBoost** ranks first (final score 9.4158), excelling in both efficiency and stability, albeit with only one win in F1-score (1.00 win count).
- **KNN** ranks a close second (9.2068). Despite winning on 5



(a) D1 CES (b) D2 CES (c) D3 CES (d) D4 CES (e) D5 CES (f) D6 CES



(g) Combined CES and Stability metrics.

Fig. 3: Detailed efficiency and stability analysis: the top row displays the dataset-specific Convergence Efficiency Score (CES) for each model; the bottom row illustrates the global trade-off between mean CES and stability across all clinical domains.

out of 6 datasets in terms of F1-score, its lower stability and slightly higher CES variability penalise it under the weighted metric.

- **AdaBoost** and **MLP** occupy the mid-tier (scores 8.0451 and 7.9275), offering decent trade-offs for applications where small energy increases are acceptable.
- **Logistic Regression** and **Random Forest** rank lowest (6.1526 and 5.6402), primarily due to instability (LR) or high carbon footprint (RF).

TABLE III: Final model ranking (Weights: $0.6 \times \mu_{\text{CES}} + 0.3 \times S + 0.1 \times W$). Final score is out of 10.

Rank	Model	μ_{CES}	Stability	Win Count	Final Score
1	XGB	10.6959	0.9032	1.00	9.4158
2	KNN	10.7781	0.6927	5.00	9.2068
3	Ada	9.6075	0.7912	0.00	8.0451
4	MLP	9.6702	0.7326	0.00	7.9275
5	LR	8.1459	0.5176	0.00	6.1526
6	RF	6.6359	0.8332	0.00	5.6402

D. Key Findings

1) **Hardware-accelerated non-parametric methods (KNN) are surprisingly green.** With `torch.cdist`, KNN achieves state-of-the-art F1-scores at near-negligible energy cost, challenging the assumption that deep models are always superior.
2) **XGBoost with CUDA histogram remains the best overall compromise:** high accuracy, excellent stability, and very low CO_2eq.
3) **Differentiable ensembles (AdaBoost) close the gap** with deep learning in accuracy but still consume more energy than tree-based or distance-based methods.
4) **Random Forest, as implemented with soft decision trees, suffers from a severe energy efficiency penalty:** a cautionary result for "green AI" benchmarking.
5) **CES successfully differentiates models** that would otherwise appear tied by F1-score alone (e.g., all models on D1 achieved 0.96 F1, but CES ranks KNN > XGB > MLP > Ada > LR > RF).

These results validate the **Convergence Efficiency Score** as a practical metric for hardware-aware benchmarking in medical machine learning.

VI. DISCUSSION

The benchmarking results across six diverse medical datasets reveal a significant paradigm shift in how diagnostic models should be evaluated. By integrating the Convergence Efficiency Score (CES) with stability analysis, we identify clear trade-offs between raw predictive power and computational sustainability.

A. The Stability-Efficiency Trade-off

Our final ranking identifies **XGBoost** as the optimal model for cross-domain clinical deployment (Final Score: 9.42). While the vectorized **KNN** implementation achieved the highest Mean CES (10.78) and the most dataset-specific "wins" (5/6 datasets), it demonstrated lower stability (7.47) compared to XGBoost (9.66). In a clinical context, this suggests that while KNN can be extremely efficient for specific data distributions, XGBoost offers a more reliable "plug-and-play" performance regardless of the medical domain or noise level. The high stability of XGBoost indicates that its gradient-boosting mechanism is less sensitive to the feature-space variances between diseases like Asthma and Heart Disease.

B. Redefining Model Hierarchy via CES

Traditional evaluation metrics often prioritize the F1-score in isolation, which tends to favor deep neural architectures like the MLP. However, the introduction of the CES metric drastically alters this hierarchy. For instance, the deep MLP achieved high accuracy in many cases but fell to Rank 4 overall due to its significant carbon footprint during the iterative optimization process. Conversely, our GPU-accelerated KNN, which is often dismissed as a "legacy" algorithm, emerged as a top-tier performer (Rank 2). This is attributed to its non-iterative nature and the massively parallelized distance computation via torch.cdist, which minimizes the active GPU telemetry window and thus maximizes energy efficiency.

C. Green AI in Clinical Diagnostics

The stability analysis proves that "Green AI" does not have to come at the cost of clinical precision. The top three models (XGBoost, KNN, AdaBoost) all maintained F1-scores above 90% while achieving high CES values. This suggests that for tabular medical data, modern boosting ensembles and optimized non-parametric models provide a more sustainable path forward than over-parameterized neural networks. By monitoring the GHG emissions specifically on the Bangladesh (BGD) energy grid, we demonstrate that regional energy profiles are a critical variable in the global "Green AI" initiative.

D. Limitations and Clinical Implications

A key limitation of the current study is the dependency on high-end NVIDIA GPU hardware for the vectorized performance of KNN and the SDT-based Random Forest. While this setup is standard in modern research hospitals, deployment in resource-constrained environments may require further quantization. Nevertheless, the high Stability Index of XGBoost suggests that it is currently the most viable candidate for a unified medical diagnostic benchmark that balances high-precision outcomes with environmental responsibility.

VII. CONCLUSION AND FUTURE WORK

A. Conclusion

In this study, we presented a high-precision, GPU-accelerated benchmarking study for medical diagnostics, evaluated across six diverse clinical datasets. The introduction of the **Convergence Efficiency Score (CES)** and the subsequent **Stability Analysis** provide a novel methodology for balancing diagnostic accuracy with environmental sustainability. Our findings demonstrate that while non-parametric models like KNN can achieve localized efficiency peaks, gradient-boosting ensembles like **XGBoost** offer the most robust and stable performance across varying clinical domains. By optimizing these models within a differentiable programming environment, we have shown that high-precision medical AI can be achieved with a minimal carbon footprint, effectively bridging the gap between "Green AI" and state-of-the-art healthcare diagnostics.

B. Future Work

Building upon the foundations of this work, several high-impact research directions are proposed:

1) **High-Order Architectures:** While this study focused on tabular clinical data, future iterations will integrate deeper architectures, including **Convolutional Neural Networks (CNNs)** for medical imaging diagnostics (e.g., MRI and X-ray analysis) and **Recurrent Neural Networks (RNNs)** or **Transformers** for time-series patient vitals and speech-based biomarker detection.
2) **Multimodal Data Integration:** We aim to expand the benchmarking suite to include non-tabular modalities such as medical speech signals and high-resolution clinical imagery, creating a unified CES benchmark for multimodal "Green AI."
3) **Edge-AI Deployment:** A critical next step involves the quantization and pruning of our top-ranked models for deployment on low-power **Edge-AI devices**. This would enable real-time, energy-efficient monitoring in remote or resource-constrained rural clinics.
4) **Hardware-Aware Optimization:** We intend to explore specialized hardware backends beyond standard CUDA, such as Tensor Processing Units (TPUs) and specialized AI accelerators, to further optimize the trade-off between floating-point operations (FLOPs) and CO_2eq emissions.
5) **Policy and Certification:** Finally, we envision the integration of the Stability Index and CES as standardized metrics for the certification of sustainable medical software, encouraging the global healthcare industry to adopt carbon-aware diagnostic tools.

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